

Institute of Business Administration
NLP with Deep Learning
Assignment 04

Due Date: 12th May 2026 (11:55 PM)

NOTE: Do not submit Google Drive or shared document links, as they can be modified after submission. The report must be submitted as a **Word or PDF** file on the **LMS**. If other files exceed the size limit, they may be uploaded via Dropbox. However, any submission via shared links will be considered incomplete and will not be graded.

This is a group assignment where the group size can be of 2-3 members. This assignment aims to provide hands-on experience with **instruction tuning** and **preference fine-tuning** of a **small base model (TinyLlama-1.1B, Qwen3-0.6B, etc.)** for instruction following and reasoning tasks. You will choose **one of two tracks (LLM or LRM)** and complete a fine-tuning pipeline with comparative analysis.

Track 1: LLM Fine-Tuning (Standard Dialogue/QA)

Preliminary:

- Select a base model.
- Generate a manual test set of prompts/questions with reference response (you may use ChatGPT, Mistral, DeepSeek to generate reference response).
- Ask base models these questions and record its response and the corresponding BLEU score +BERTScore. Note: Your ChatGPT/Clause/Gemini response would serve as the gold answer.

Choose Option A or B

Option A – Standard Pipeline (SFT → DPO)

1. Supervised Fine-Tuning (SFT)
 - Dataset: Any instruction–response dataset from HuggingFace (not used in class).
 - Perform five fine-tuning trials total. In each trial, vary both LoRA parameters (rank, target modules) and training arguments (learning rate, batch size, epochs). Collectively, your five trials should explore a wide range of configurations.
 - Selecting the Best SFT Model: After completing your 5 SFT trials, select one to proceed to Step 2 using:
 1. BLEU + BERTScore on the 10 prompts that you tested with baseline model earlier.
 2. If scores turn out to be similar, then use Lower validation loss as the tie breaker.
 - Report BLEU + BERTScore for each trial and clearly state which trial was selected as best and why.
2. Preference Fine-Tuning (DPO)
 - Dataset: Any preference dataset (chosen, rejected responses).

- Fine-tune the best SFT model from step 1 using DPO + LoRA.
- Report 5 DPO trials (vary beta, LR, batch size, epochs).
- Select the best SFT+DPO model using:
 1. BLEU + BERTScore on the 10 prompts that you tested with baseline model earlier.
 2. If scores turn out to be similar then use Lower validation loss as the tie breaker.
- Report BLEU + BERTScore for each trial and clearly state which trial was selected as best and why.

Option B – Multi-Dataset SFT (No DPO)

- Select two different instruction datasets (e.g., Alpaca + Dolly).
- Perform SFT separately on each dataset, i.e., run 5 trials on Dataset A, then 5 trials on Dataset B (10 total). Compare best from each. Note: Do **not** fine-tune sequentially (D1 → D2), as this would confound the contributions of each dataset.
- Compare performance of both SFT models against the base model (as outlined in Option A – SFT).
- Discuss your findings in terms of why one data set performed better than the other.

Track 2: LRM Fine-Tuning (Reasoning-Focused)

Preliminary:

- Select a base model.
- Generate a manual test set of prompts/questions with reference response (you may use ChatGPT, Claude or Gemini to generate reference response). Make sure that the question/response requires reasoning.
- Ask base models these questions and record its response. Use an LLM as a judge (not the same used for generating sample response) to evaluate reasoning quality. So if you generated sample response from ChatGPT then don't use ChatGPT as a Judge and instead use DeepSeek, or Mistral or Kimi, etc.

Choose Option C or D:

Option C – SFT → GRPO

1. SFT on reasoning data
 - Select any reasoning dataset from Hugging face.
 - Perform 5 SFT trials with LoRA.
 - Selecting the Best SFT Model: After completing your 5 SFT trials, select one to proceed to Step 2 using:
 1. Scores from the LLM-as-a-Judge on the 10 prompts that you tested with baseline model earlier.
 2. If scores turn out to be similar then use Lower validation loss as the tie breaker.
 - Report LLM-as-a-Judge score for each trial and clearly state which trial was selected as best and why.
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2. GRPO (Group Relative Policy Optimization)

- Fine-tune best SFT model using GRPO.
- Report 5 GRPO trials (vary group size, KL coeff, LR).
- Select the best SFT+GRPO model using:
 1. Scores from the LLM-as-a-Judge on the 10 prompts that you tested with baseline model earlier.
 2. If scores turn out to be similar then use Lower validation loss as the tie breaker.
- Report LLM-as-a-Judge score for each trial and clearly state which trial was selected as best and why.

Option D – Multi-Dataset Reasoning SFT

1. Select **two reasoning datasets** from Hugging Face.
2. Perform SFT separately on each dataset, i.e., run 5 trials on Dataset A, then 5 trials on Dataset B (10 total). Compare best from each. Note: Do **not** fine-tune sequentially (D1 → D2), as this would confound the contributions of each dataset.
3. Compare performance of both SFT models against the base model (as outlined in Option C – SFT on reasoning data).
4. Discuss your findings in terms of why one data set performed better than the other.

Deliverables

1. Submit clean and well-documented Python code notebook(s) that were used for experimentation via GitHub or as a zipped folder.
2. A well-organized report, either in the docx or PDF format.

Report Guidelines

Organize your report into clearly defined sections. In your report, include discussion on:

1. **Platform Details:** Specify the platform used for experimentation (e.g., local machine, Kaggle, Colab). If multiple platforms were used, clarify where each stage was executed.
2. **Data Details:** Clearly state the sources of the datasets, including the size and number of samples used in the dataset. You may use subsets of these datasets as appropriate for your compute budget but justify your choices in the report. Describe the datasets selection and preprocessing steps
3. **Experimentation, Analysis, and Insight:** Clearly document the experimental setup and results, highlighting insights gained from multiple trials.
 - Model and tokenizer choice
 - Evaluation metrics
 - The impact of different LoRA and training configurations on the responses.
 - Differences in behavior between:
 - Base model
 - Instruction-tuned model
 - Preference-tuned model (after DPO or GRPO)
 - List parameters of the best-performing models (e.g., LoRA rank, DPO beta, learning rate).
 - Output quality: include examples that illustrate improvement or degradation

- Resource usage and training time
 - Strengths and weaknesses of SFT DPO and GRPO (for respective options/experiments), the impact of hyperparameter choices, and scenarios where each approach excels.
 - Common failure cases or unexpected behaviors.
4. **Reproducibility:** Your report must provide enough detail to enable others to replicate your work. Include any information that is critical for reproduction, such as preprocessing steps, system configuration, or model fine-tuning techniques.

Additional Instructions:

- **Application Development (Optional):** If you have developed a working application based on your system, include screenshots and relevant details in an appendix at the end of your report.
- **Figures and Tables:** Ensure that all figures and tables are properly numbered and cited in the text. Avoid vague references like “the figure below”; instead, use precise citations such as “Table 1 shows...” or “As shown in Figure 7...”.
- **References:** If you have used external resources, such as blogs or GitHub repositories, ensure they are appropriately cited. Include a reference section before the appendix to acknowledge all sources and avoid any potential issues of plagiarism. Proper citation is a key part of your academic and professional training.
- **Submission File Name:** The file name should be as per the group members name and don't name it Assignment1 or Project1. So, if there are two group members, Aamna and Zaid, then name it Aamna_Zaid.docx.